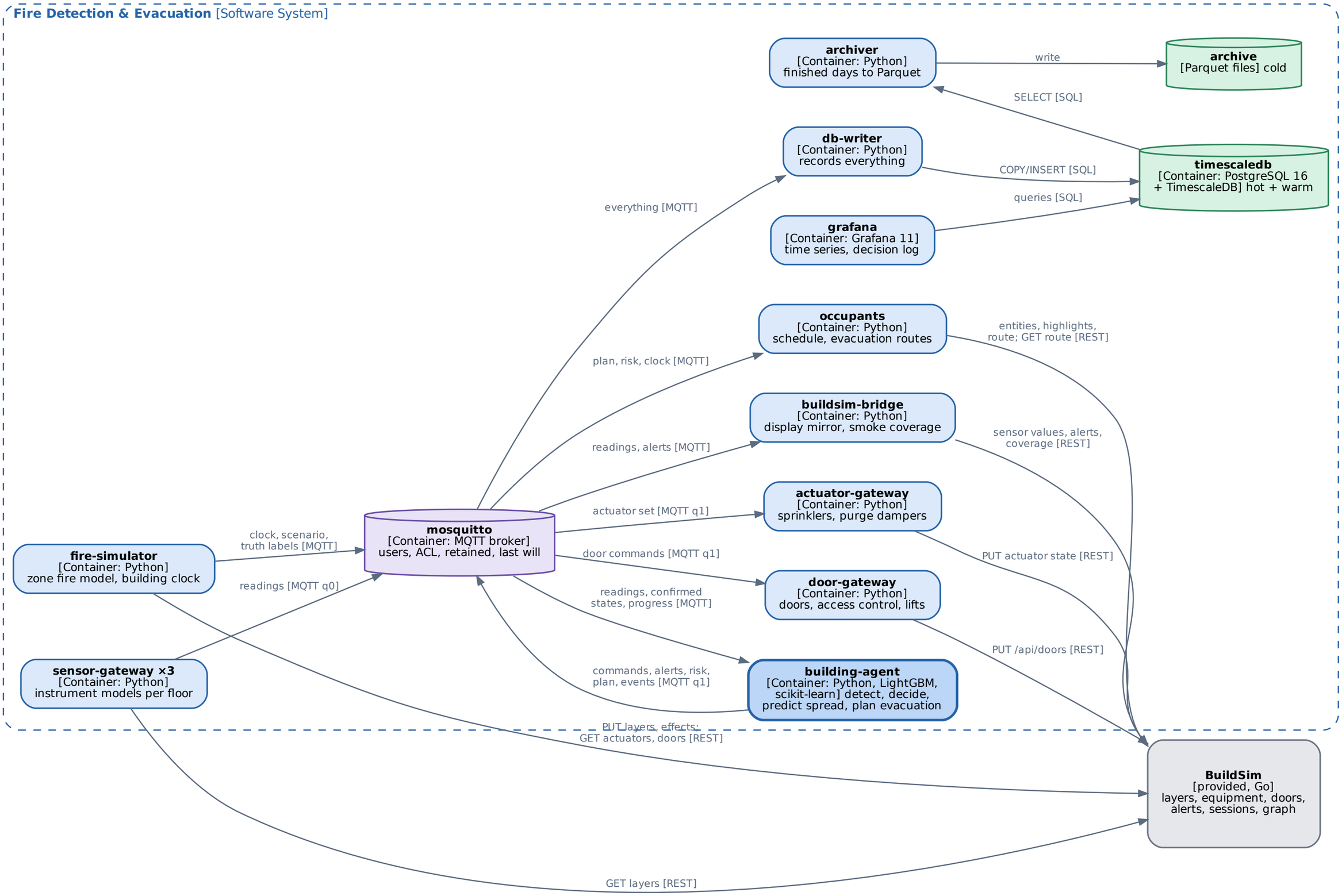
Fire Detection & Evacuation [Software System]
archiver
[Container: Python]
finished days to Parquet
write
archive
[Parquet files] cold
SELECT [SQL]
db-writer
[Container: Python]
records everything
COPY/INSERT [SQL]
timescaledb
[Container: PostgreSQL 16
+ TimescaleDB] hot + warm
queries [SQL]
everything [MQTT]
grafana
[Container: Grafana 11]
time series, decision log
occupants
[Container: Python]
schedule, evacuation routes
entities, highlights,
route; GET route [REST]
plan, risk, clock [MQTT]
buildsim-bridge
[Container: Python]
display mirror, smoke coverage
sensor values, alerts,
coverage [REST]
readings, alerts [MQTT]
actuator-gateway
[Container: Python]
sprinklers, purge dampers
actuator set [MQTT q1]
clock, scenario,
truth labels [MQTT]
mosquitto
[Container: MQTT broker]
users, ACL, retained, last will
door commands [MQTT q1]
PUT actuator state [REST]
fire-simulator
[Container: Python]
zone fire model, building clock
readings [MQTT q0]
door-gateway
[Container: Python]
doors, access control, lifts
readings, confirmed
states, progress [MQTT]
PUT /api/doors [REST]
sensor-gateway ×3
[Container: Python]
instrument models per floor
commands, alerts, risk,
plan, events [MQTT q1]
building-agent
[Container: Python, LightGBM,
scikit-learn] detect, decide,
predict spread, plan evacuation
PUT layers, effects;
GET actuators, doors [REST]
BuildSim
[provided, Go]
layers, equipment, doors,
alerts, sessions, graph
GET layers [REST]